Exact Two-Dimensional $SU(2)$ Yang–Mills in Lean: Weyl Integration, Heat-Kernel Convolution, Migdal Invariance, and the Exact Simple-Loop Area Law

Lluís Eriksson

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Abstract

We give a kernel-checked Lean 4/Mathlib formalization of the exact heat-kernel solution of two-dimensional $SU(2)$ Yang–Mills theory along the chain from Haar measure to a concrete Migdal subdivision move. The development identifies the transported Haar probability on $SU(2) \simeq S^3$ with the canonical spherical probability, proves the required all-order orbital integration law, derives translated character convolution for every pair of irreducible labels, passes from finite character sums to the infinite heat-kernel semigroup by dominated convergence, and integrates an actual edge shared by two faces. A further structural induction now eliminates every internal edge of a finite positive-area polygonal disk cellulation whose dual graph is a tree, proves that the procedure uses exactly one fewer integration than the number of faces, and produces a nontrivial concrete instance of the project's exact-area-law interface. Combined with all-label coefficient extraction, this yields the exact normalized simple-loop identity

$$\mathbb{E}[W_n(C)] = \exp\left[-\frac{n(n+2)}{4}\,\mathrm{Area}(C)\right].$$

All displayed endpoints compile without `sorry`, `admit`, or project-local axioms. The mathematical identities are classical; the claimed contribution is their concrete end-to-end machine-checked composition, not a new analytic solution of two-dimensional Yang–Mills theory.

1 Statement and claim boundary

Let dg be normalized Haar probability on $SU(2)$, let χ_n be the character of the irreducible representation of dimension $d_n = n+1$, and put

$$c_n = \frac{n(n+2)}{4}, \qquad K_t(g) = \sum_{n\geq 0} d_n e^{-tc_n}\chi_n(g), \qquad t > 0.$$

With convolution

$$(f * h)(g) = \int_{SU(2)} f(x)h(x^{-1}g)\,dx,$$

the formalized chain is

$$\begin{aligned} &\text{Haar on } S^3 \Longrightarrow \text{all-order orbital law} \Longrightarrow \chi_n * \chi_m \\ &\qquad \Longrightarrow K_s * K_t = K_{s+t} \Longrightarrow \text{Migdal} \Longrightarrow \text{simple-loop coefficient.} \end{aligned}$$

This result is distinct from the programme's earlier strong-coupling area-law theorem. The earlier theorem is a volume-uniform *inequality* for finite lattices and $SU(N_c)$ in an explicit convergence window, obtained from a Kotecký–Preiss cluster expansion. The present theorem is an *exact identity* in the two-dimensional $SU(2)$ heat-kernel model, obtained by Haar integration and subdivision. The former is cited context; it is not used to discharge any exact two-dimensional statement here.

No claim is made here about four-dimensional Yang–Mills, a continuum construction in four dimensions, or a new proof of the classical exact solution. A priority claim about formalization should remain conditional on a systematic literature and repository search.

2 From spherical Haar measure to the orbital law

Write an element of $SU(2)$ in its standard form

$$g = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Earlier modules construct a homeomorphism $SU(2) \simeq S^3$, realize left multiplication as an ambient real-linear isometry, prove preservation of the canonical `Measure.toSphere` measure, and use uniqueness of normalized Haar measure to obtain literal equality of the two probability measures.

The new all-order step studies $u = |a|^2$. In dimension four, the squared mass in one complex rail is uniform on $[0, 1]$. Rather than assume this distribution, the formal proof establishes its moment recurrence, proves

$$\int_{SU(2)} u^k \, dg = \frac{1}{k+1} \qquad (k \geq 0),$$

and then identifies measures on the compact interval using polynomial density. The resulting literal endpoint is:

```
theorem su2FirstRailMassMeasure_eq_uniform :
  su2FirstRailMassMeasure = su2UnitIntervalMeasure
```

For $x, y \in [-1, 1]$, choose representatives with half-traces x, y and average their product over conjugation. The preceding uniform law reduces the orbital integral to a one-variable Chebyshev integral. For the Chebyshev polynomial U_m of the second kind, the formalized formula is

$$\int_{SU(2)} U_m\left(\tfrac{1}{2}\operatorname{tr}(kak^{-1}b)\right) dk = \frac{U_m(\frac{1}{2}\operatorname{tr} a) U_m(\frac{1}{2}\operatorname{tr} b)}{m+1}.$$

The proof includes the endpoint cases $x = \pm 1$; no division by a vanishing sine is hidden in the statement.

3 All-order translated character convolution

Characters are implemented concretely as

$$\chi_n(g) = U_n\left(\tfrac{1}{2}\operatorname{tr} g\right).$$

Conjugacy invariance, left and right Haar invariance, Fubini, and the orbital law give the translated orthogonality relation

$$\int_{SU(2)} \chi_n(x)\chi_m(x^{-1}g) \, dx = \delta_{nm} \frac{\chi_n(g)}{n+1}. \tag{1}$$

This is stronger than orthogonality at the identity and is exactly the consumer needed by the heat-kernel product.

Theorem 3.1 (Machine-checked character convolution)**.** *For every $n, m \in \mathbb{N}$ and $g \in SU(2)$,*

$$(\chi_n * \chi_m)(g) = \begin{cases} \chi_n(g)/(n+1), & n = m, \\ 0, & n \neq m. \end{cases}$$

The public Lean endpoints are listed below. They are theorems about the actual normalized Haar integral, not fields projected from an abstract character package.

```
integral_su2CharacterReal_mul_translate
su2CharacterChebyshev_convolution
```

4 Infinite heat-kernel semigroup

For a cutoff N, expand both finite kernels, interchange the finite sums and the Haar integral, and apply (1). The Kronecker selector leaves a single sum and the Casimir weights multiply:

$$e^{-sc_n} e^{-tc_n} = e^{-(s+t)c_n}.$$

This proves the finite convolution identity. The infinite theorem then uses the already established uniform Casimir majorant for positive times. Partial kernels converge pointwise and uniformly; the product is bounded by the product of two summable majorants, so dominated convergence passes the Haar integral to the limit.

Theorem 4.1 (Concrete heat semigroup)**.** *For $s, t > 0$ and every $g \in SU(2)$,*

$$\int_{SU(2)} K_s(x) K_t(x^{-1}g)\, dx = K_{s+t}(g).$$

Equivalently, as functions, $K_s * K_t = K_{s+t}$.

The corresponding Lean declarations, including the function-level consumer, are:

```
su2HeatKernelPartial_convolution
su2HeatKernel_convolution
su2HeatKernel_convolution_consumer
```

5 A concrete Migdal move

Consider two adjacent faces of areas s and t. Let a, b encode the external boundary holonomies and integrate their shared oriented edge x. The actual two-face density in the development is

$$D^{a,b}_{s,t}(x) = K_s(ax) K_t(x^{-1}b).$$

Left invariance changes variables $y = ax$ and the heat semigroup gives

$$\int_{SU(2)} D^{a,b}_{s,t}(x)\, dx = K_{s+t}(ab). \tag{2}$$

Theorem 5.1 (Two-face Migdal subdivision)**.** *Equation (2) holds for all $a, b \in SU(2)$ and all $s, t > 0$. It is proved by a genuine Haar integration over the shared edge and invokes the concrete infinite heat-kernel semigroup.*

This closes the first nontrivial subdivision move: two holonomy variables, two faces, and one integrated shared edge. The endpoints below contain no package hypothesis.

```
su2Migdal_twoFace_merge
su2Migdal_subdivision_invariant
```

6 Finite planar reduction and termination

The local move can now be iterated over a nontrivial global class. A `SU2PlanarFaceTree` is generated from positive-area elementary faces by binary gluing along one oppositely oriented boundary edge. Geometrically this describes polygonal disk cellulations with tree dual. If C has left and right subcellulations L, R, its amplitude is defined recursively by

$$A_C(a,b) = \int_{SU(2)} A_L(a,x) A_R(x^{-1}, b)\, dx.$$

Thus the definition contains one actual Haar variable for every internal edge; orientation is encoded by the occurrences x and x^{-1}.

The face count $F(C)$, internal-edge count $E(C)$, and total area $T(C)$ are structural recursions. Lean proves

$$E(C) = F(C) - 1, \qquad T(C) > 0,$$

and then eliminates all internal variables by induction on C:

$$A_C(a,b) = K_{T(C)}(ab). \tag{3}$$

At each gluing node the induction hypotheses expose two heat kernels and the proof invokes the concrete two-face theorem. Hence termination is supplied by the inductive type itself and the exact number of integration steps is certified separately by $E = F - 1$.

The explicit collapsed cellulation has one face, no internal edge, and area $T(C)$. A consumer theorem proves that its amplitude is literally equal to the original recursively integrated amplitude. The principal declarations are:

```
SU2PlanarFaceTree.internalEdgeCount_eq_faceCount_sub_one
su2PlanarCellulationAmplitude_eq_heatKernel
su2PlanarCellulationAmplitude_eq_effectiveFace
```

For each representation label n, the formalization defines a plane-loop theory whose loops are these finite cellulations, whose area is $T(C)$, whose string tension is $n(n+2)/4$, and whose Wilson functional integrates the full recursive amplitude. Equation (3) followed by coefficient extraction gives a nontrivial concrete instance `su2TreePlanarExactAreaLawPackage`. The generic public interface is then consumed by `su2TreePlanar_simpleLoop_areaLaw_exact`; no field of that package remains assumed. The trivial-label case also proves the normalization

$$\int_{SU(2)} A_C(1,g)\, dg = 1.$$

7 Exact simple-loop coefficient

The same repository proves, for every label n and positive area t, that the normalized Wilson character $W_n = \chi_n/(n+1)$ extracts the corresponding heat coefficient:

$$\int_{SU(2)} W_n(g) K_t(g)\, dg = e^{-tc_n} = \exp\left[-t\frac{n(n+2)}{4}\right].$$

The proof first establishes the identity for partial character sums and then passes to the full kernel by dominated convergence. Its final consumer is:

```
su2_exact_simpleLoop_areaLaw
```

Together with the iterated Migdal reduction, this is the exact two-dimensional route for tree-dual polygonal disks: subdivision does not change the density, every internal edge is eliminated, and the remaining effective face has the Casimir coefficient prescribed by its total area.

8 Audit and reproducibility

The checked source is built with Lean 4 and Mathlib. The public root module imports the concrete orbit, character-convolution, heat-semigroup, and exact area-law modules. A repository-wide placeholder audit finds no occurrence of `sorry`, `admit`, or a project-local `axiom` declaration in the Lean development. Printing axioms for the twenty audited public endpoints returns only the standard kernel dependencies

`propext`, `Classical.choice`, `Quot.sound`.

The toolchain and Mathlib revision are pinned in the repository. GitHub Actions builds the public root target from a clean checkout and separately rejects placeholders.

Principal new declarations.

```
su2FirstRailMoment_recurrence
su2FirstRailMoment_eq
su2FirstRailMassMeasure_eq_uniform
intervalIntegral_chebyshevU_product_formula
integral_su2FirstEntryMass_eq_unitInterval
integral_su2Haar_orbit_character_general
integral_su2CharacterReal_mul_translate
su2CharacterChebyshev_convolution
su2HeatKernelPartial_convolution
su2HeatKernel_convolution
su2Migdal_twoFace_merge
su2Migdal_subdivision_invariant
su2_exact_simpleLoop_areaLaw
SU2PlanarFaceTree.internalEdgeCount_eq_faceCount_sub_one
su2PlanarCellulationAmplitude_eq_heatKernel
su2PlanarCellulationAmplitude_eq_effectiveFace
su2TreePlanar_wilsonExpectation_eq_casimir
su2TreePlanarExactAreaLawPackage
su2TreePlanar_simpleLoop_areaLaw_exact
integral_su2PlanarCellulationAmplitude_eq_one
```

9 Scope and next frontier

The present artifact closes the compact-group analytic chain and a global finite reduction for polygonal disk cellulations whose dual graph is a tree. It does not yet formalize arbitrary embedded planar graphs with interior vertices or cycles in the dual graph, gauge fixing on a general cellular decomposition, nonsimple or intersecting loops, continuum measure construction, or higher-genus partition functions.

The next mathematically meaningful extension is therefore precise: define a general finite planar combinatorial map, choose and verify an elimination or gauge-fixing schedule in the presence of dual cycles, and prove independence of that choice. The tree-dual theorem must be used as a base case, not relabeled as the arbitrary-planar result.

References

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[2] E. Witten, *On quantum gauge theories in two dimensions*, Commun. Math. Phys. **141** (1991), 153–209.

[3] B. K. Driver, *Yang–Mills theory and the heat equation on compact Lie groups*, J. Funct. Anal. **110** (1992), 1–34.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, Lecture Notes in Computer Science 12699 (2021), 625–635.

[5] The Mathlib Community, *The Lean mathematical library*, CPP 2020, 367–381.